

Vinoth Mani

Python Backend Engineer • New Jersey • 8573505054 • vinoth.mani@outlook.com • LinkedIn • GitHub • Portfolio

Summary

Backend software engineer with 7+ years of experience in Python backend development, microservices architecture, and AWS cloud platforms, delivering high-availability, production-ready systems across multiple domains in Agile environments.

Technical Skills

Languages & Backend: Python, REST API, Flask, FastAPI, Django

Cloud & DevOps: AWS (EC2, S3, RDS, CloudWatch, AWS SAM, SES, API Gateway, Code service, Lambda, CloudFormation)

Database: SQL, DynamoDB, NoSQL, Firebase, Redis

CI/CD & Tools: Docker, GitHub Actions, Jenkins, Git, Keygen, pytest, VS code, postman, Meld, Jira, curl, Confluence, Packer

Libraries: SQLAlchemy, PyQt, RASA, NLU, Nuitka, NumPy, Pandas

Practices: Microservices, System Design, Agile, Object-Oriented Programming, Automation, SDLC

Certifications

- AWS Certified Solutions Architect – Associate.
- RASA AI Developer.
- OpenEDG Certified Python Professional.

Work Experience

Capgemini (Johnson & Johnson Inc), New Jersey

Senior Consultant

Nov 2024 - Present

- Design and develop backend microservices and REST APIs using Python FastAPI, Flask for enterprise applications.
- Implement automation for build, test, and deployments using GitHub Actions and Jenkins, reducing manual effort and improving release reliability.
- Build and deploy containerized services using Docker across AWS environments.
- Design and maintain AWS-based backend infrastructure, supporting secure and scalable production systems.
- Perform code reviews, lead technical discussions, and provide backend best-practice guidance.
- Actively manage deployments, production support, and issue resolution, collaborate directly with client stakeholders to translate business requirements into technical solutions.

Blockalytics, Atlanta, GA

Software Engineer

Feb 2024 – Nov 2024 | Jul 2022 – Dec 2022

- Led Flask-based REST API development, improving system performance and frontend integration.
- Designed and developed Python microservices, containerized with Docker and deployed on AWS EC2 via CloudFormation, with Route 53 for DNS and CloudWatch for monitoring to ensure scalability and fault tolerance.
- Built and enhanced CI/CD pipelines using GitHub Actions and AWS Code services, improving software stability by 89%, reducing deployment risks, and shortening lead times.
- Engineered Python-based automated build system incorporating API modules, achieving a 99% reduction in build time.
- Implemented automated API testing with pytest and strengthened software integrity through Nuitka-based binary builds, maintaining 100% software quality standards.
- Developed and published a custom Python library to Gemfury, integrating Keygen-based licensing and using NumPy and Pandas for outage detection and advanced system analysis.

Scientific Games, Chennai, India

Associate Software Engineer

Sep 2017 - Aug 2021

- Orchestrated a 28% user base expansion for a casino platform through advanced JavaScript enhancements, leveraging Leaflet.js and TypeScript to elevate gaming experiences and market competitiveness.
- Achieved a notable 19% UI improvement in poker and slot games, optimizing performance by deploying on Amazon Web Services utilizing services Route53, CloudWatch, Cloud Formation, EC2, VPC, IAM.
- Configured middleware components in Django for handling various aspects of request/response processing.
- Revolutionized game development by optimizing build systems in Python, and shell scripts, achieving a 100% improvement in OS compatibility and build time.
- Improved efficiency and scalability by streamlining workflows, introducing an augmented reality treasure hunt feature in gaming solutions, with a strategic initiative, yielding a substantial 30% improvement in overall code quality.
- Resolved issues and introduced a 6% faster card shuffling feature, enhancing UI rendering and functionality.

Education

Northeastern University, Boston, MA

Dec 2023

Master of Science, Software Engineering Systems

Anna University, Chennai, India

Apr 2018

Bachelor of engineering, Computer Science Engineering